QQ-C-581a

April 4, 1967

SUPERSEDING
Fed. Spec. QQ-C-581
May 31, 1932

FEDERAL SPECIFICATION

COPPER, SILICON (ALLOYING ADDITIVE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers three silicon copper alloys.

1.2 Classification. Silicon copper shall be furnished in the following alloys and forms, as specified (see 6.2).

Alloys:
- A—11 percent silicon
- B—20 percent silicon
- C—30 percent silicon

Forms:
- Shot
- Lumps
- Ingots
- Notched slabs

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein:

Federal Specifications:

QQ-C-521 —Copper Ingot.
PPP-B- 41—Barrels; Wood, Slack.
PPP-B-585—Boxes, Wood, Wirebound.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-D-705—Drum: Metal Shipping, Steel. (Over 12 and Under 55 Gallon)
PPP-D-723—Drums, Fiber.

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151—Metals: Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 9630

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Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129—Marking for Shipment and Storage.
MIL–STD–147—Palletized and Containerized Unit Loads 40" X 48" 4 Way (Partial) Pallet Skids, Runners, or Pallet-Type Base.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) Standard:

B 53—Specification for Silicon Copper.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material. Silicon copper shall be made using copper in accordance with QQ–C–521, with or without the addition of clean scrap resulting from the manufacture of silicon copper of similar composition.

3.2 Chemical composition. The chemical composition of silicon copper shall be as shown in table I.

TABLE I. Chemical composition, percent

Element	Alloy A	Alloy B	Alloy C
Silicon	10.0 – 12.0	18.5 – 21.5	28.5 - 31.5
Tin	0.25 max.	0.25 max.	0.25 max.
Zinc	.25 max.	.25 max.	.25 max.
Iron	.50 max.	.50 max.	.50 max.
Aluminum	.25 max.	.25 max.	.25 max.
Calcium	.15 max.	.15 max.	.15 max.
Copper	Rem.	Rem.	Rem.
Copper + silicon + iron	99.4 min.	99.4 min.	99.4 min.

3.3 Form. Silicon copper shall be furnished in the form of notched slabs or ingots of such size that they can be conveniently handled. When specified (see 6.2), shot or lumps shall be furnished.

3.4 Identification marking. The manufacturer's name or brand and melt number be legibly and indelibly marked on each slab, ingot and container, by which the manufacturer and the heat may be identified.

3.5 Workmanship. All forms of silicon copper shall be commercially clean and free from deleterious foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspecion. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection lot. A lot shall consist of all silicon copper of the same alloy and form submitted for inspection at one time, but shall not exceed 1,000 pounds.

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4.3 Sampling.

4.3.1 For chemical analysis. A sample for chemical analysis shall be taken from each lot. It shall consist of not less than two ounces of material removed from the samples selected in accordance with ASTM B 53.

4.3.2 For visual examination. From each lot of material, samples shall be selected for visual examination in accordance with MIL-STD-105 at inspection level I. The acceptable quality level (AQL) shall be 2.5 percent defective.

4.4 Visual examination. Each sample selected in accordance with 4.3.2 shall be subjected to surface examination for workmanship requirements.

4.4.1 Inspection of preparation for delivery. Examination for the preservation, packaging, packing, and marking for shipment shall be made for conformance with the requirements of section 5.

4.5 Test procedures.

4.5.1 Chemical analysis. The samples selected in accordance with 4.3.1 shall be analyzed in accordance with method 111 or 112 of Fed. Test Method Std. No. 151. In case of dispute, referee analysis shall be by method 111.

4.6 Rejection.

4.6.1 Examination defects. Any sample unit having one or more defects shall be rejected. If the number of nonconforming sample units in the sample exceeds the acceptance number specified in 4.3.2 for the sample size, the entire lot shall be rejected subject to the provisions of the section on "Acceptance and Rejection" of MIL-STD-105.

4.6.2 Test failures. A lot shall be rejected for failure to meet any one of the test requirements when tested in accordance with 4.5, subject to the provisions of the section on "Rejection and Retests" of Fed. Test Method Std. No. 151.

5. PREPARATION FOR DELIVERY

5.1 Packing (see 6.2).

5.1.1 Level A.

5.1.1.1 Slabs and ingots. Unless otherwise specified (see 6.2), slabs and ingots shall be packed separately in a box conforming to PPP-B-601, overseas type; or PPP-B-621, class 2; or PPP-B-585, class 3. The gross weight of each box shall not exceed 200 lbs. The boxes shall be strapped in accordance with the appendix to the applicable box specification. When specified (see 6.2), slabs and ingots shall be prepared for shipment on pallets conforming to MIL-STD-147.

5.1.1.2 Lump and shot. Lump and shot of like size and description shall be packed, in quantities not to exceed 50 pounds, in fiber drums conforming to PPP-D-723, type II; or in metal drums conforming to PPP-D-705; or in wood barrels conforming to PPP-B-11.

5.1.2 Level B.

5.1.2.1 Slabs and ingots. Slabs and ingots shall be packed as specified in 5.1.1.1 except the containers shall be domestic type and class and the gross weight shall not exceed the weight limitations of the applicable box specification. When specified (see 6.2), the slabs and ingots may be shipped loose without additional packing.

5.1.2.2 Lump and shot. Lump and shot of like size and description shall be packed as specified in 5.1.1.2 except that fiber drums may be type I.

5.1.3 Level C. Silicon copper shall be prepared for shipment to insure carrier acceptance and safe delivery to destination in accordance with rules and regulations applicable to the mode of transportation.

5.2 Marking (see 6.2 and 6.3).

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5.2.1 Civil agencies. In addition to markings required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to markings required by the contract or order, shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Intended use. Silicon copper is used as a vehicle for the addition of silicon in the melting and preparation of copper alloys.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Alloy and form required (see 1.2, 3.2 and 3.3.).
(c) Size of ingots (see 3.2.1 and 3.3).
(d) Selection of applicable level of packing required (see 5.1).
(e) Whether slabs and ingots are to be shipped loose (see 5.1.2.1).
(f) Preparation for shipment, if other than specified (see 5.1).

6.3 The packaging, packing, and marking requirements specified herein apply to direct shipment to the Government, and are not intended to apply to contracts or orders between the manufacturer and the prime contractor.

6.4 Related specification. This specification covers the same material as ASTM B 53.

MILITARY CUSTODIANS:

Army—MR
Navy—OS
Air Force—69

Review activities:

Army—MR, MU, WC
Navy—OS
Air Force—69

User activities:

Army—None
Navy—SH
Air Force—None

CIVILIAN AGENCIES INTEREST:

COMM
JUS
AGRI
GSA
VA

Preparing activity:

Army—MR

Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification tobtain extra copies of documents referenced herein. Price 5 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119 R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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